

What is chemistry?

THE SCIENCE THAT DEALS WITH THE PROPERTIES OF SUBSTANCES AND LOOKS AT HOW THEY CAN CHANGE FROM ONE TYPE TO ANOTHER.

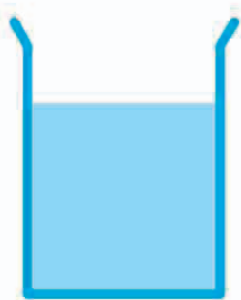
Chemistry is sometimes called the central science, because it forms the link between physics and biology. Chemistry builds on the knowledge of physics and then, in turn, is used to provide the basis for much of biology.

Understanding substances

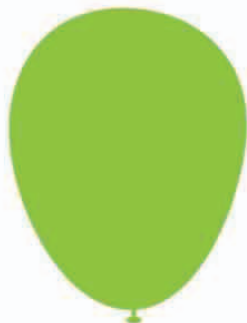
Chemists seek to understand how the characteristics and structure of a substance, natural or artificial, can be described. What is it about water that makes it a flowing liquid, while the plastic bucket used to carry it is a rigid solid? A chemist finds the answer at the very smallest of scales. Every substance contains atoms, and the way they are arranged dictates how a substance behaves.

▽ Describing materials

Chemists have many ways of describing substances. They include the substance's state—solid, liquid, or gas—or whether it is metallic like the screw, or nonmetallic like the seashell.



water—liquid, nonmetallic



helium in balloon—gas, nonmetallic



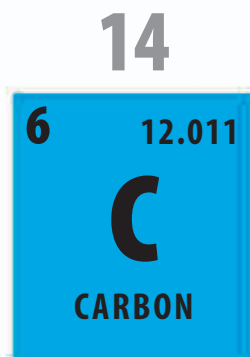
seashell—solid, nonmetallic



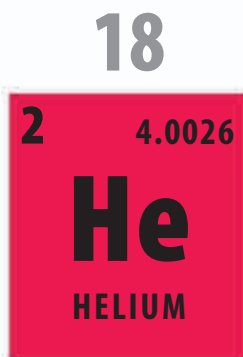
screw—solid, metallic

Elements

Everything in the Universe is made out of raw materials called elements. There are about 91 naturally occurring elements. Most, such as gold or mercury, are pretty rare, while others, such as carbon, chlorine, and iron, are found in great quantities. Few elements are found pure in nature; they are usually combined with other elements to form entirely different materials called compounds. (Water is a compound of hydrogen and oxygen, for example.) Compounds can be separated into their elements, but an element cannot be broken down into anything simpler.



carbon element



helium element

◁ Defining elements

Chemists arrange the elements in the periodic table (see pages 116–117) according to the structure of their atoms. For example, the number of electrons (one of the parts of an atom) at a certain part of the atom means carbon is in group 14 whereas helium is in group 18.